

Supplementary Materials for *Children, but not language models, show accelerating returns in word learning*

Materials and Methods

Data and code

Reproducible code for this paper is available at <http://github.com/langcog/standard-model-2>. Data were retrieved from Wordbank using the `wordbankr` package (1) and CHILDES using the `childesr` package (2).

All longitudinal datasets with three or more observations per child were included. We applied a quality filter to exclude administrations that showed either decreases of more than 25% of productive vocabulary or increases that exceeded 40% of total vocabulary per month. This excluded 36 children in total (1.9%), nearly all from the English (Marchman) dataset (16%; Norwegian 0.5%, none from the others) (SI: Data Exclusions).

Bayesian models

Using Stan (3), Bayesian models were fit separately to each dataset per the specification described in the text. The model was parameterized to capture individual children’s variation from the population mean such that $\xi_i = \mu_\xi + a_i$ and $\kappa_i = \mu_\kappa + b_i$, where $(a_i, b_i) \sim \text{MVN}(\mathbf{0}, \Sigma)$. To facilitate interpretation of the accumulator mechanism, we included the constant $\log H = \log(365) \approx 5.90$ (waking hours/month) and the anchor age $a_0 = 18$ months (so $\log(t/a_0) = 0$ at 18 months).

We chose weakly informative priors throughout, including on the efficiency and acceleration parameters $\mu_\xi \sim \mathcal{N}(-6, 5)$ and $\mu_\kappa \sim 1 + \mathcal{N}(0, 5)$ (centered around the pure accumulator model) as well as the individual child standard deviations $\sigma_a \sim \text{Half-Normal}(0, 3)$ and $\sigma_b \sim \text{Half-Normal}(0, 5)$ and the item difficulties $\tau_\delta \sim \text{Half-Normal}(0, 5)$. We also chose a prior that mildly regularized the correlations between a_i and b_i ($\mathbf{L} \sim \text{LKJ}(2)$).

All models were fit using `cmdstanr`, with four independent chains of 1,000 warmup and 1,000 sampling iterations each (4,000 post-warmup draws) and a target acceptance rate of 0.9. The primary series of models was additionally refit using 2,000 warmup and 8,000 sampling iterations. We assessed convergence using rank-normalized split- $\hat{R}$ and bulk effective sample size. For parameters we interpret (primarily κ , σ_a , σ_b), we required $\hat{R} < 1.01$ and bulk ESS above 400.

Learning efficiency

Age of acquisition for each word was computed based on the weighted average of word difficulties δ_j from the three fitted M3 models for English. For the population level of ξ and κ , we computed the

earliest age at which there was at least a .5 probability of producing the word. Average exposures per month to each word were computed as $N_j = R \cdot t_{50}(j) \cdot p_j$, where R is the rate of input (words/month), $t_{50}(j)$ is the word-level age of acquisition computed above, and p_j is the word’s corpus probability. We used average input estimates derived from (4) and (5) for our estimate of R (see SI: Input Estimation), and computed p_j from empirical word frequencies in all North American English corpora in chldes-db release 2021.1.

Language model training

Following (6), we fit randomly-initialized GPT-2-small (124M parameters) models, using the 24M-word (~47M-token) preparation of English-language CHILDES data from that study. Consistent with the previous work, we included children’s own utterances in the training data. Children do in fact hear their own speech; more importantly, removing one side of every interaction would create disjoint, unnatural training data. We also fit models to the 2nd BabyLM challenge dataset (downloaded from osf.io/ad7qg, with CHILDES data removed) (7) and the ClimbMix dataset (available at <https://huggingface.co/datasets/karpathy/climbmix-400b-shuffle>) (8). We constructed three disjoint subsets of 24M words from each.

For comparability, all data were tokenized using a standard BPE tokenizer trained on the full CHILDES dataset (all 24M). Surprisal was computed as the mean per-word negative log likelihood over ~50 held-out validation contexts (last 128 tokens each) for the 609 CDI words that were used in (9) and that appeared in the training data with sufficient frequency in the CHILDES held-out contexts. Surprisal values for sigmoid fits were read out from the epoch with the lowest validation loss.

For the computation of surprisal across training runs (“training”), we trained models using 3 random seeds on the full dataset. For the computation across different dataset sizes (“development”), the data were segmented into 18 nested datasets ranging in size from .5M to the full 24M. We then chose 10 random seed values and trained 10 models on each dataset for a total of 180 models. For the computation of variability across model runs (“composition”), we ran 8 random seed values across 12 nested dataset sizes for three disjoint subsets of both BabyLM and ClimbMix, resulting in 8 (seeds) x 12 (sizes) x 3 (subsets) x 2 (datasets) = 576 distinct models.

Models were trained for a total of 20 epochs with sequence length 1024 and batch size 8, using AdamW for optimization and learning rate of 1e-4 with a linear decay, no warmup, weight decay of 0, and no in-epoch shuffling. Training was performed on NVIDIA A40 GPUs. All trained models are available at <https://huggingface.co/mcxfrank/chldes-gpt2-ladder> and <https://huggingface.co/mcxfrank/gpt2-composition-control>.

Proportional return on input

Proportional return on input was computed for models based on mean surprisal $L(D)$ of the CDI words after training on D word tokens of input. This follows the form from (10) of $L(D) = E + B D^{-\beta}$, with E the irreducible entropy floor and B, β fit across budgets (here $E \approx 2.94$ nats, $B \approx 312$, $\beta \approx 0.32$, $R^2 = 0.998$). The excess loss $L(D) - E = B D^{-\beta}$ is the reducible part that training removes; it falls as a power law of slope $-\beta$. Marginal return is the loss removed per e -fold of data,

$$R(D) = -\frac{dL}{d \ln D} = \beta (L(D) - E),$$

obtained by differentiating the scaling law. We compute the ratio of this quantity to the excess loss, obtaining the proportional return: the share of the remaining reducible loss removed per e -fold, $\gamma_{\text{LM}}(D) = \frac{R(D)}{L(D)-E} = \beta$. For a power law, this is constant: $\gamma_{\text{LM}} = \beta \approx 0.32$ at every budget, recapitulating the scaling law.

We can compute the same quantity for children. We define a vocabulary production loss, $L_i(t) = \text{mean}_j[-\log p_{ij}(t)]$, whose floor is zero (a known word is produced, $p \rightarrow 1$). Its marginal return is $R_i(t) = -dL_i/d \log t = \kappa_i \text{mean}_j(1 - p_{ij})$, so the proportional return is

$$\gamma_i(t) = \frac{R_i(t)}{L_i(t)} = \kappa_i \frac{\text{mean}_j(1 - p_{ij})}{\text{mean}_j(-\log p_{ij})},$$

which rises toward the child's own κ_i as the vocabulary saturates.

Comparison with Prior Models

Our accelerating accumulator model has not been described in the literature in the exact form we use, but it is closely related to other models. Here we describe these models as special cases of our model. We write the latent ability of child i on word j at age t as

$$\eta_{i,j}(t) = \log r_i + \log \alpha_i + \kappa_i \log t - \delta_j,$$

where r_i and α_i decompose the term ξ_i in our model: r_i is the input rate for the child and α_i is their learning efficiency. Several prior accumulator models can then be seen as special cases in which one or more of these components is fixed:

- (11) is a pure accumulator in which $\kappa = 1$ and children do not differ from one another ($\sigma_\alpha = \sigma_\kappa = 0$), so both input rate and efficiency reduce to a constant, corresponding to our M0.
- (12) introduces a model in which ability grows as t^{D+1} , creating acceleration. This exponent is exactly our κ , and this model is equivalent to our M1.
- (13) present our M2: they allow α to vary across individuals, but not κ .

In addition, several other models in the literature can be described as variants of this framework:

- (14) augment the pure M0 accumulator with a “leverage” term, in which words learned earlier facilitate the learning of later words. Their central result is a negative one: leverage changes the shape and timing of vocabulary growth, but it cannot create acceleration that is not already present in the distribution of word difficulties.
- (15) define a pure accumulator model with $\kappa = 1$, but with an additional parameter relating to the speed of per-word accumulation. This model maps onto a combination of our M0 and the 2-parameter logistic variant that we fit below.

Non-identifiability in Cross-Sectional Data

Since acceleration is a property of change over time, a single cross-sectional observation from an individual child cannot identify the child’s acceleration. As fig. S1 shows, a single observation from a child could be accounted for by variation in either ξ_i (efficiency) or κ_i (acceleration).

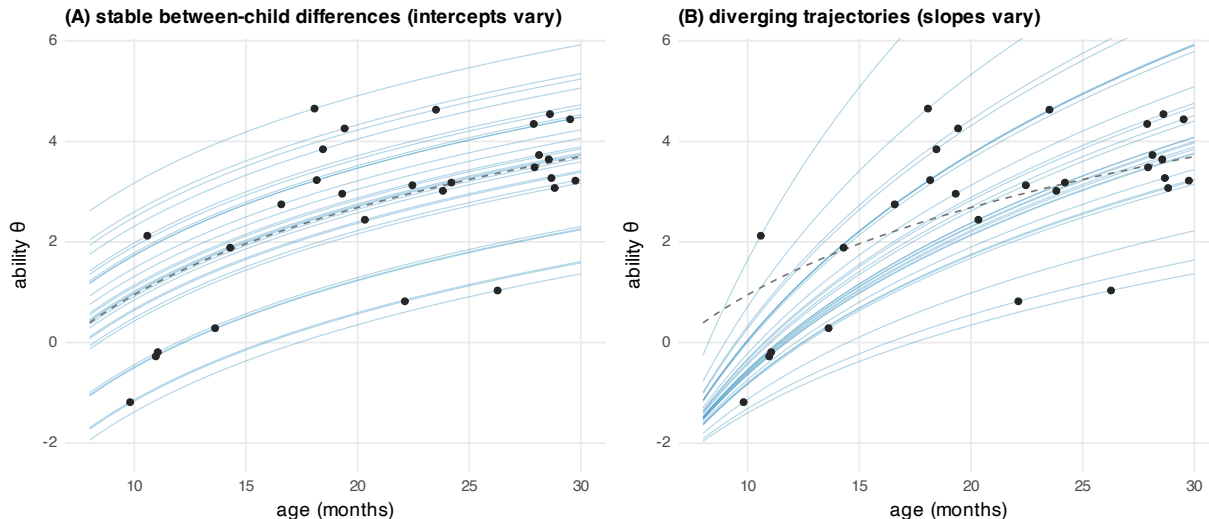


fig. S1: Non-identifiability of growth models in cross-sectional data. Panels show the same simulated data, reflecting the possibility of stable between-child intercept differences (A) vs. varying growth slopes (B).

Datasets

table S1 gives the characteristics of each dataset used. In the main text we focus on children with three or more administrations (for more precise estimates of κ), but below we also confirm the robustness of our claims including children with two or more administrations.

table S1: Characteristics of all datasets used in the paper. N and ages reflect the analysis samples (post-exclusion). Adm/Chi gives the median administrations per child.

Citation	Language	N (3+/2+)	Admins	Age (months)			Adm/Chi
				Mean	Min	Max	
Thal (Wordbank)	English (American)	639 / 653	1,919	19.0	12	29	3
Smith (Wordbank)	English (American)	152 / 316	607	22.2	16	30	4
Marchman (Wordbank)	English (American)	162 / 2,136	499	20.3	8	30	3
Simonsen et al. (2014)	Norwegian	792 / 1,630	4,770	24.1	8	36	5
Hagihara et al. (2023)	Japanese	96 / 187	289	16.5	11	33	3

fig. S2 shows the full spaghetti plots for each dataset, highlighting in red those observations that failed our quality check and were excluded. We speculate that these observations are due to either data entry issues or faulty correspondences between observations. Such issues occur frequently in archival datasets and affect only a small portion of our data.

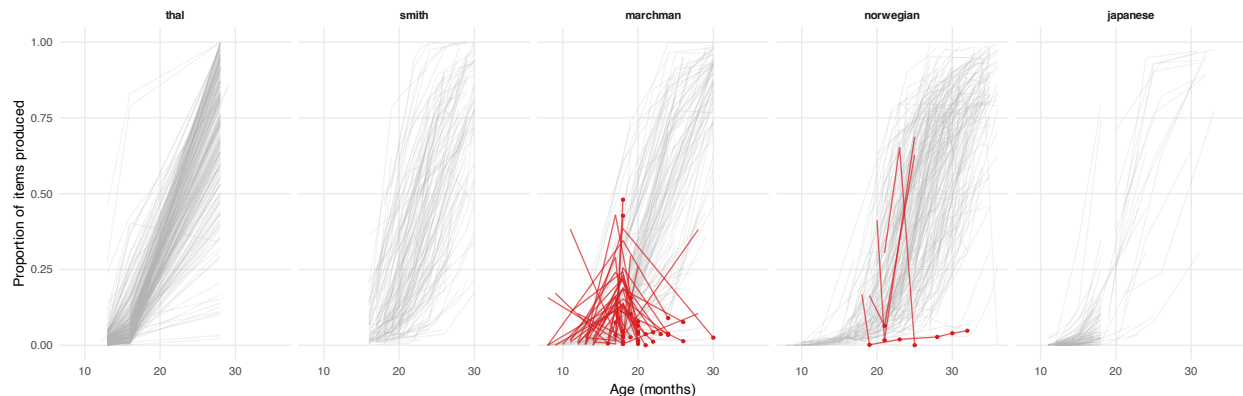


fig. S2: Data-quality exclusions. Red lines show children excluded from the analysis entirely, with red dots marking flagged administrations; gray lines are a random sample of retained children, shown for reference.

Full Model Comparison Results

table S2 gives the full leave-one-out model comparison results across models for both the dataset with 3+ administrations/child and the larger dataset with 2+ administrations/child.

table S2: Bayesian leave-one-out (LOO) model comparison. Each cell is the difference in expected log predictive density (ELPD) relative to the best model (M3), with the standard error of the difference in parentheses; more negative is worse. Every step of the ladder improves fit in every dataset, and the full model M3 (per-child efficiency *and* acceleration) is preferred throughout. Built from the per-fit LOO objects (paper/build_cache_short.R).

Model	Δ ELPD vs M3 (SE)				
	Thal	Smith	Marchman	Norwegian	Japanese
≥ 3 administrations (main text)					
M0. accumulator ($\kappa = 1$)	-154,678 (350)	-102,675 (340)	-83,954 (285)	-130,396 (369)	-24,628 (186)
M1. + acceleration	-46,214 (289)	-62,282 (309)	-40,868 (258)	-70,888 (336)	-7,923 (126)
M2. + efficiency var.	-7,880 (133)	-6,542 (121)	-4,878 (110)	-5,702 (111)	-761 (40)
M3. + acceleration var.	0	0	0	0	0
≥ 2 administrations (full longitudinal)					
M0. accumulator ($\kappa = 1$)	-155,002 (351)	-120,701 (372)	-125,833 (355)	-131,793 (373)	-53,239 (260)
M1. + acceleration	-45,613 (287)	-75,934 (343)	-66,770 (332)	-75,942 (346)	-21,922 (197)
M2. + efficiency var.	-7,723 (132)	-7,795 (129)	-14,004 (195)	-6,047 (115)	-1,919 (65)
M3. + acceleration var.	0	0	0	0	0

Individual-level Cross-validation

The model comparisons above report ELPD LOO values for held-out data from individual item-responses, which are nested within children. To assess the value of M3 (the accelerating accumulator) in making fully out-of-sample predictions, we designed a prospective test. As our data, we selected a subset of the Norwegian sample that included 371 children with ≥ 6 longitudinal CDIs. We then fit models to the child's most recent k observations and used these to predict the child's next

administration (3 months later). This design avoids confounding data density with the gap between the fitted and predicted observations.

Results from this analysis are shown in table S3. The result is very sensitive to the value of k used. With $k = 2$, the fixed slope model (M2) clearly predicts better because the per-child slope model (M3) fully parameterizes the two observations per child and absorbs measurement error as a result. This effect decreases as k increases until the two models are tied by $k = 5$. We expect that with even denser data, a separate slope parameter would lead to individually better fits, but in practice the single slope model makes acceptable predictions.

Importantly, because both M2 and M3 contain a population acceleration parameter, this analysis does not test for the presence of acceleration. Instead it suggests that measuring the precise amount of acceleration each child is undergoing requires sustained, precise longitudinal measurement.

table S3: Prospective cross-validation for Norwegian children with at least six administrations ($n = 371$). Each row describes a model fit to k administrations and scored on the child’s held-out final sitting. dELPD/child is the per-child difference in expected log predictive density, full model minus shared-slope model; negative favors the shared slope. sigma_b is the between-child acceleration SD in the training fit, against a full-data value of 5.65.

k	horizon (mo)	$\Delta\text{ELPD}/\text{child}$	SE	z	% children better	σ_b
2	3	-32.41	6.77	-4.8	39	12.56
3	3	-5.32	4.10	-1.3	49	7.86
4	3	-0.33	3.64	-0.1	48	6.20
5	3	+0.02	3.51	+0.0	51	5.17

We next tested whether acceleration plays an identifiable role in predicting individual children’s trajectories at all. We therefore fit $M2_0$, a variant of M2 with the acceleration exponent constrained to the pure accumulator value ($\kappa = 1$) — rather than the population acceleration rate — while retaining the per-child efficiency term ξ_i . The comparison of M2 and $M2_0$ thus tests the hypothesis that acceleration is predictive of individual trajectories. table S4 shows the results of this comparison. M2, the accelerating model, predicted held-out administrations better at every depth.

table S4: Prospective cross-validation of acceleration, on the same Norwegian children and held-out administrations as table S3. M2 retains a free population acceleration exponent; M2-zero removes it. dELPD/child is the per-child difference in expected log predictive density, M2 minus M2-zero; positive favors acceleration.

k	horizon (mo)	$\Delta\text{ELPD}/\text{child}$	SE	z	% children better
2	3	+91.87	6.97	+13.2	84
3	3	+143.21	8.90	+16.1	88
4	3	+197.27	10.33	+19.1	90
5	3	+248.10	11.22	+22.1	92

Comparison to a Non-Parametric Quantile Model

To test whether the accelerating accumulator appropriately fits the distribution of growth curves across children, we compared against a strong baseline: the flexible non-parametric GAMLSS family that is used to produce normative curves for the CDI (16). Specifically, for each dataset we fit a GAMLSS beta-regression of the proportion of words produced by the child’s age. We used a penalized spline function on the mean and on the scales.

fig. S3 shows that across all five datasets the M3 fit (blue) tracks both the non-parametric GAMLSS fit (orange) and the empirical quantiles (circles). Thus, the accelerating accumulator is a faithful, low-dimensional compression of the data’s quantile structure.

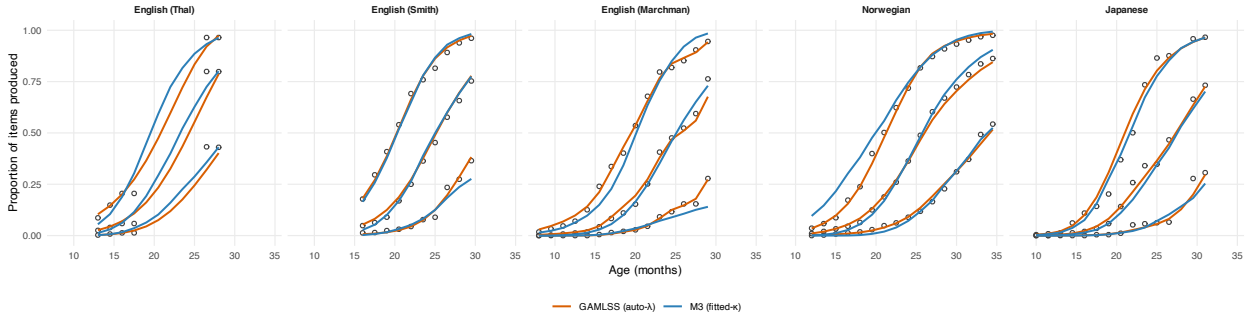


fig. S3: M3 fits (blue, drawn from fitted per-child acceleration) vs. non-parametric GAMLSS beta-regression (orange), with empirical 10th/50th/90th percentiles (open circles). Lines are the 10/50/90 percentiles of each model. Color denotes model.

Robustness Across Model and Data Settings

In the main text, we report fits to children with 3+ administrations with separate models fit to each dataset, but our parameter estimates were substantially similar across two other settings: first, separate fits to children with 2+ administrations, and second, a full hierarchical model fit to children with 3+ administrations. table S5 shows LOO fits and the robustness of the κ estimates across these different settings.

The pooled model was structured similarly to the by-dataset models, but children and words were modeled as nested within their own dataset. Each dataset had its own mean ξ and κ values. Dataset-level standard deviations on efficiency and acceleration (σ_a and σ_b) were distributed as $\log \sigma_{a[d]} \sim \mathcal{N}(m_a, s_a)$ and $\log \sigma_{b[d]} \sim \mathcal{N}(m_b, s_b)$, so a dataset with few children had its estimated spread drawn toward the typical value across datasets, while a dataset with many children was left essentially unconstrained. The item-difficulty distribution was estimated separately within each dataset, since the different datasets reflect data from different languages. Overall, the pooled model reproduced the parameters of the independent fits closely.

table S5: For each dataset and setting: number of children (N), population acceleration (κ), between-child acceleration SD (σ_b), and the per-observation LOO advantage of the full model M3 over M2 (ΔELPD per item response).

Setting	N	κ	σ_b	$\Delta\text{ELPD}/\text{obs}$ (M2→M3)
English (Thal)				
≥2 separate	653	11.5	3.19	0.0154
≥3 separate (main text)	639	11.5	3.19	0.0158
pooled (≥3)	639	11.5	3.22 [3.07, 3.38]	—
English (Smith)				
≥2 separate	316	12.8	8.06	0.0156
≥3 separate (main text)	152	12.9	5.14	0.0159
pooled (≥3)	152	12.8	5.13 [4.67, 5.63]	—
English (Marchman)				
≥2 separate	2,136	10.6	6.89	0.0280
≥3 separate (main text)	162	10.6	6.60	0.0164
pooled (≥3)	162	10.4	6.51 [5.87, 7.30]	—
Norwegian				
≥2 separate	1,630	12.9	7.63	0.0121
≥3 separate (main text)	792	13.3	5.65	0.0114
pooled (≥3)	792	13.2	5.65 [5.40, 5.93]	—
Japanese				
≥2 separate	187	12.0	5.46	0.0074
≥3 separate (main text)	96	11.6	5.45	0.0059
pooled (≥3)	96	11.5	5.40 [4.74, 6.25]	—

Sensitivity to Data-Quality Exclusions

To ensure that the exclusion filter we used in the main text does not result in significant changes to our estimates, we refit the model at four settings: with the filter disabled entirely, at a loose threshold, at the reported threshold, and at a tight one (table S6). These refits apply only to two datasets, English (Marchman) and Norwegian. The exclusion filter does not result in major changes to κ estimates in either case.

table S6: Population acceleration and between-child acceleration SD by data-quality filter setting, for the two datasets in which the filter removes observations. The loose filter removes children with a decline of more than 40% below a child’s running peak or a rise of more than 60 percentage points per month; the main filter uses a decline of 25% and a rise of 40%, as reported in the Methods; and the tight filter uses 15% and 25%.

Filter	Removed	κ	σ_b
English (Marchman)			
none	0	9.37	8.33
loose	29	11.17	8.30
main	32	10.65	6.60
tight	34	10.44	5.55
Norwegian			
none	0	13.05	5.77
loose	36	13.19	5.64
main	46	13.25	5.65
tight	55	13.22	5.34

Linear vs. Logarithmic Age

We additionally fit M3 variants with linear, rather than logarithmic, growth in acceleration over time. table S7 shows that the linear model was dispreferred across all five datasets.

table S7: Log-age vs linear-age accumulation. LOO ELPD advantage of the log-age model (M3) over its linear-age counterpart, with SE; positive favors log-age.

Dataset	Δ ELPD (log – linear)
Thal	267 (31)
Smith	249 (23)
Marchman	281 (26)
Norwegian	323 (39)
Japanese	39 (8)

2PL Model Comparison

In the main analysis we use a Rasch (1PL) item model, in which every word has a difficulty but all words have equal discrimination between children. (17) found that a 2PL model, which adds a per-word discrimination λ_j , was preferred over the 1PL for CDI data. We therefore refit M3 with a per-word discrimination parameter.

table S8 shows the results of this comparison. The 2PL does genuinely fit better, with substantial differences in LOO ELPD for every language. However, model results are otherwise unchanged. Estimates of κ increase somewhat, but they are not directly comparable across item models. In normal IRT models, ability scores are constrained to a standard normal distribution, but in our model, abilities grow over time so there is no single standard scale, and the addition of the discrimination parameter can change the overall scale. For this reason, we report $\log \kappa$, since a pure change of units would be a constant shift. In addition, across individual words, we find that the predicted age

of acquisition (the age at which the word is predicted to have a 50% probability of being known) is closely aligned, suggesting that the models are highly comparable.

Critically, δ (difficulty) and λ (discrimination) correlations were small to modest. If harder words were substantially more discriminating, this correlation could inflate estimates of κ in the 1PL model; in practice we find little support for this hypothesis.

table S8: Rasch (1PL) versus two-parameter logistic (2PL) item models, M3 refit on the same data. Discrimination: SD of $\log \lambda_j$, its 10th-90th percentile range on the natural scale, and its correlation with item difficulty (90% interval). Acceleration: population κ under each item model and the difference in $\log \kappa$. Age of acquisition: correlation and median absolute difference in per-word t_{50} , which is invariant to the discrimination scale. LOO: expected log predictive density of the 1PL relative to the 2PL, with SE; negative favors the 2PL.

Dataset	Discrimination			Acceleration κ			Word slopes $\lambda_j \kappa_i$		Age of acquisition		LOO
	SD $\log \lambda$	λ p10–p90	$r(\lambda, \delta)$	1PL	2PL	$\Delta \log \kappa$	med.	% > 1	$r(t_{50})$	med. $ \Delta t_{50} $	ΔELPD (SE)
English (Thal)	0.25	0.73–1.34	+0.13 [+0.11, +0.15]	11.5	12.4	+0.072	12.2	99.8	0.999	0.13	-2,897 (82)
English (Smith)	0.27	0.71–1.37	+0.10 [+0.07, +0.13]	12.9	13.7	+0.053	12.9	99.3	0.996	0.13	-2,726 (79)
English (Marchman)	0.24	0.74–1.33	+0.05 [+0.02, +0.09]	10.6	11.6	+0.089	11.1	98.4	0.998	0.25	-1,412 (59)
Norwegian	0.29	0.70–1.36	+0.22 [+0.21, +0.23]	13.3	14.1	+0.065	13.7	99.4	0.995	0.16	-4,426 (108)
Japanese	0.34	0.65–1.51	+0.07 [+0.02, +0.12]	11.6	15.3	+0.279	14.7	100.0	0.986	1.47	-641 (41)

Sensitivity to Item Selection

One potential concern about our κ estimates is that they could be influenced by the composition of the CDI instrument. Could a broader spread of item difficulties result in a larger κ estimate? To address this issue, we refit M3 on the same children and same administrations, but with the item set narrowed. We fit models under three sets of conditions: refitting the middle 50% and middle 25% of items (to narrow the width of the distribution), refitting the easiest and hardest halves separately (to shift the difficulty distribution), and subsetting randomly to 50% and 25% of items. If κ scales as a function of the difficulty distribution (σ_δ), recovered κ values should scale with the standard deviation of the retained items; if not, κ should be relatively invariant. fig. S4 shows the results of this analysis. Overall, κ values were invariant to retained item variability.

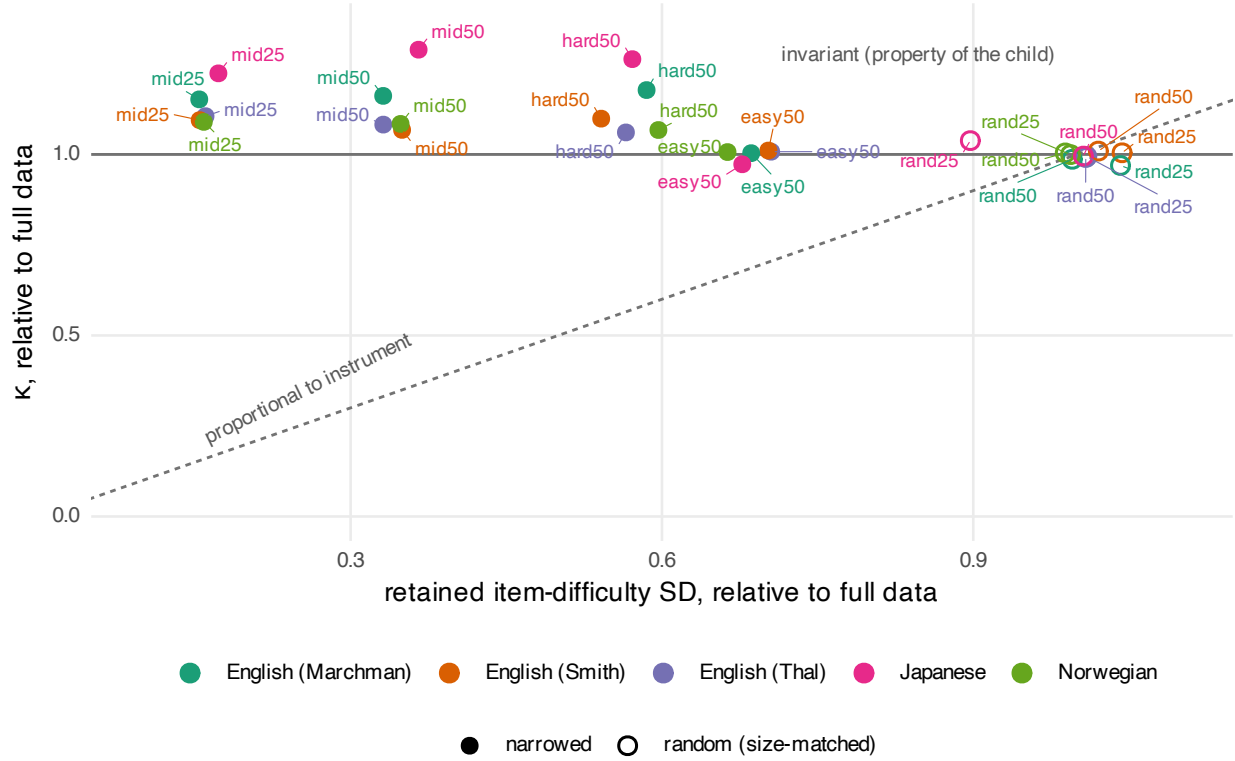


fig. S4: Acceleration under deliberately narrowed item sets, plotted as the ratio to the full-data fit against the ratio of retained item-difficulty spread. Each point is one refit on the same children and administrations. If acceleration rescaled with the instrument, points would fall on the diagonal; if it is a property of the children, they fall on the horizontal line at one. Open points are random subsets of matched size, which discard the same number of items without narrowing the spread.

Variation in Acceleration

To explore the distribution of individual children's fitted parameters, we show the best linear unbiased predictor (BLUP) for each child (fig. S5) as well as the random effects of the fitted models (table S9). While the distribution appears generally normal, we do not give a strong interpretation as this shape may in part be due to the normal form of the model's random effect structure. However, across all datasets, acceleration appeared right-skewed.

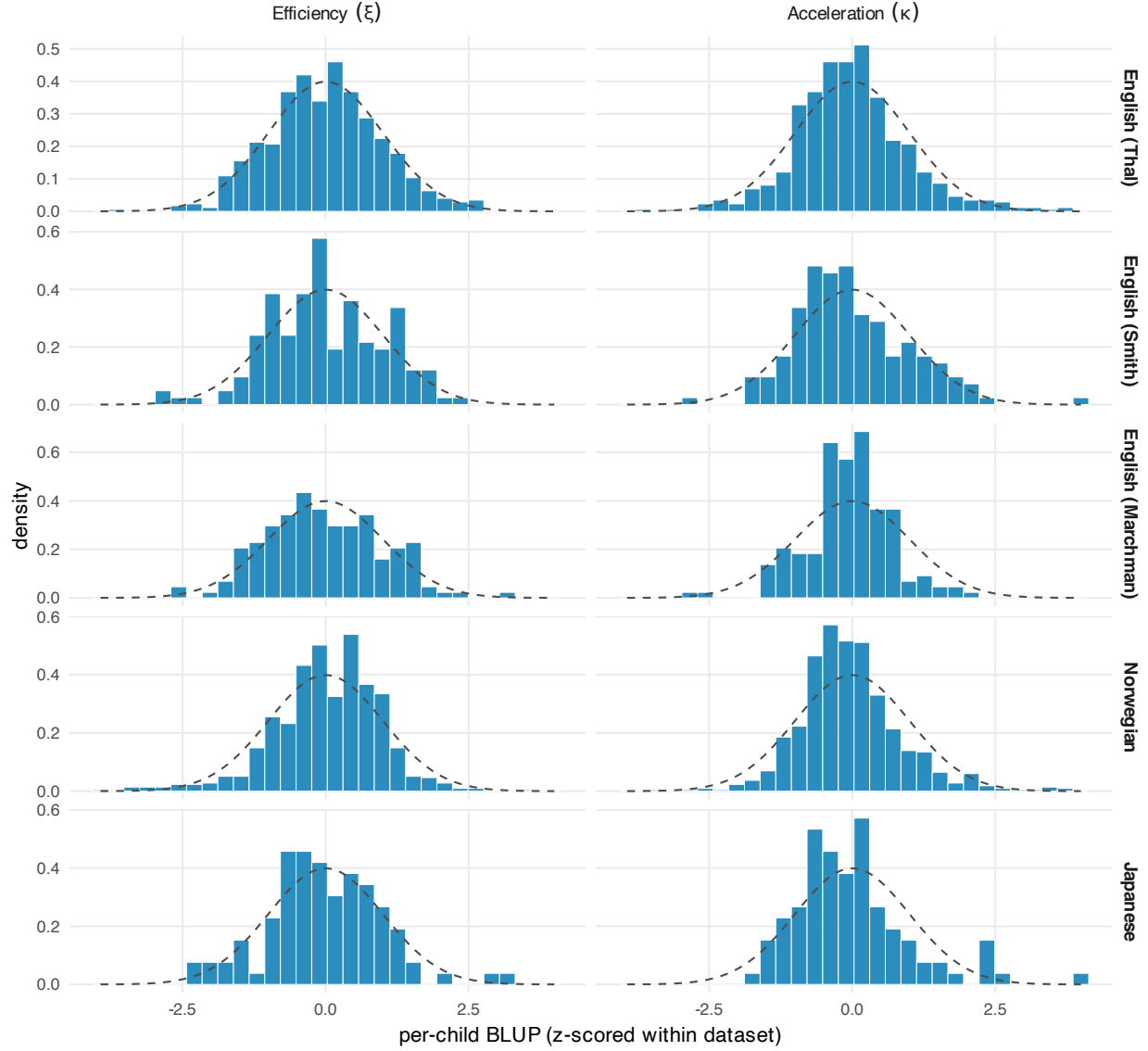


fig. S5: Histograms of per-child BLUP estimates, z-scored within each dataset, for both efficiency and acceleration parameters. Dashed lines indicate the standard normal reference distribution.

table S9: Random-effect parameters for M3 fits, by dataset: population acceleration κ , between-child SD in efficiency (σ_a) and acceleration (σ_b), their correlation (ρ), and the percentage of children estimated above the pure-accumulator value $\kappa = 1$. Brackets give 90% posterior intervals.

Dataset	N	κ	σ_a	σ_b	ρ	% $\kappa_i > 1$
English (Thal)	639	11.5 [11.5, 11.6]	1.56 [1.49, 1.64]	3.19 [3.05, 3.35]	+0.10 [+0.03, +0.17]	99.8
English (Smith)	152	12.9 [12.8, 13.1]	2.08 [1.89, 2.31]	5.14 [4.69, 5.68]	-0.26 [-0.39, -0.13]	99.3
English (Marchman)	162	10.6 [10.5, 10.8]	1.80 [1.65, 1.99]	6.60 [5.90, 7.32]	+0.17 [+0.04, +0.29]	97.5
Norwegian	792	13.3 [13.2, 13.3]	2.71 [2.60, 2.83]	5.65 [5.41, 5.88]	-0.62 [-0.66, -0.58]	99.5
Japanese	96	11.6 [11.1, 12.1]	1.95 [1.71, 2.25]	5.45 [4.74, 6.32]	+0.26 [+0.07, +0.43]	100.0

Input Estimation

We make use of estimates of children’s language input in both **?@fig-efficiency** and **?@fig-llm-acceleration**, panel C. We calculate average language input per month by estimating words per hour and multiply by an estimated number of waking hours per month. Word tokens per hour estimates come from 42 children with manually transcribed data in two prior studies (4, 5, estimates used also in 13). Across these studies, the pooled geometric mean is $\exp(7.338) \approx 1,537$ word tokens per hour of child-directed (any-adult) speech, with between-child standard deviation of log rate $\sigma_r = 0.534$. We multiply these estimates by an assumed 12 waking exposure-hours per day to get a monthly average number of hours ($12 \times 30.4 \approx 365$), giving $1,537 \times 365 \approx 561\text{K}$ tokens per month. A child’s average cumulative input at age t months is then $t \times 561\text{K}$ under a constant-rate assumption.

In **?@fig-llm-acceleration**, panel C, we give uncertainty bands on representative children based on ± 1 SD of the between-child log input rate. A child one standard deviation above (below) the median rate reaches the same age having heard $\times 1.71$ ($\div 1.71$) as many tokens. This results in a horizontal shift of the child’s curve along the log-token axis.

Our calibration computations assume that input is constant across age. To provide empirical verification, we used data from four pre-existing datasets: (18), (19) (BabyView), (20), (21) (SEEDLings). These studies measured children’s input using either LENA recorders or head-mounted cameras; we computed total words heard per hour in each dataset for each recording. fig. S6 shows speech rate per hour in these datasets. Slopes over age for each were shallow and non-significant.

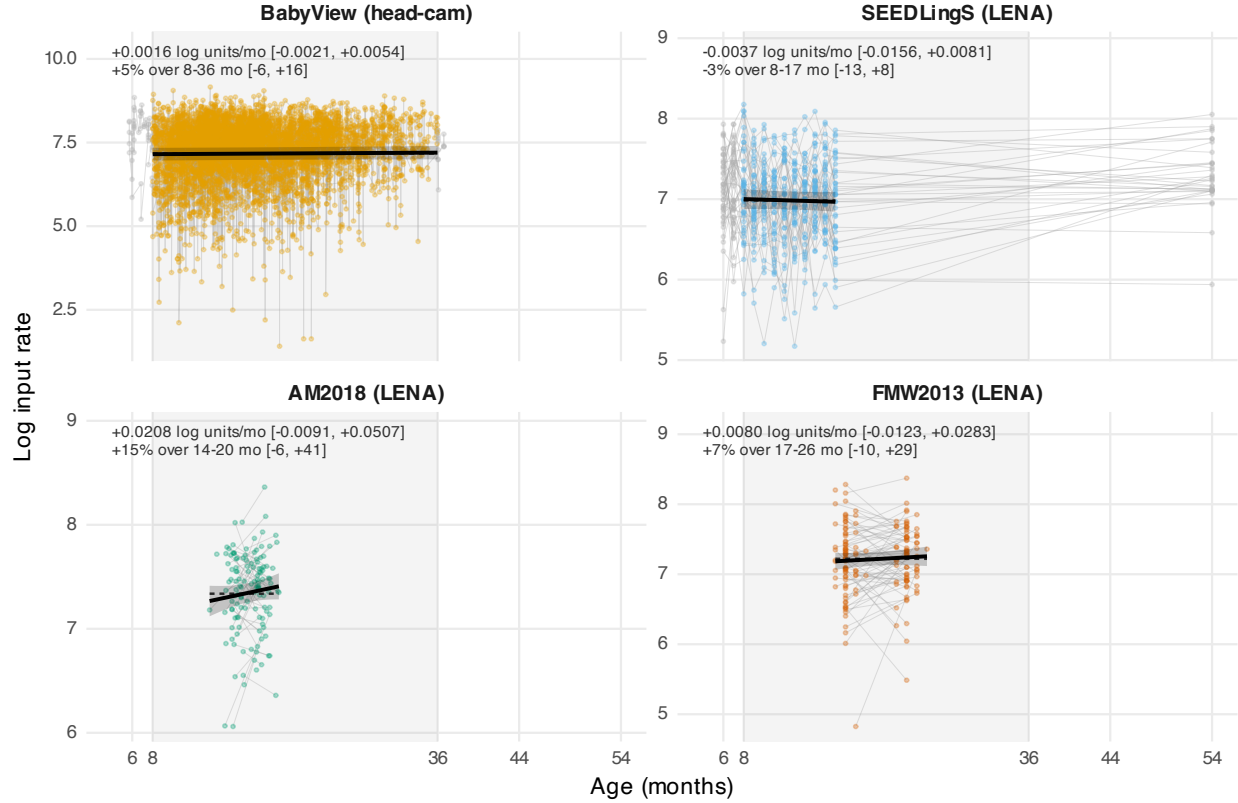


fig. S6: Input rate does not change appreciably across development. Each panel plots log adult input rate against child age, with one point per recording, faint gray lines joining recordings from the same child, and the fitted within-child trend (black) with its 95% confidence band. Dashed lines show zero slope. Trends are fitted over 8-36 months (shaded), the range over which the constant-rate assumption is applied; gray points fall outside that window and are excluded from the fit.

Age of Acquisition in the Pure Accumulator Model

Figure 2 in the main text converts each word's fitted difficulty into an estimated number of cumulative exposures at the age it is acquired. Since later-acquired words are also rarer, here we ask whether that frequency change would be sufficient to produce the observed relationship between cumulative exposure and age of acquisition.

fig. S7 shows the same relationship, but comparing age of acquisition estimates between M3 (the accelerating accumulator) and M0 (the pure accumulator). The pure accumulator does not reproduce either the scale or the slope of the relationship (table S10). Within every word class, the slope of the relationship is substantially shallower in M0 than M3, and the M0 forces the fitted difficulties to spread acquisition across 5 to 254 months, against 14 to 35 months for the accelerating model and 8 to 36 months of actual observation.

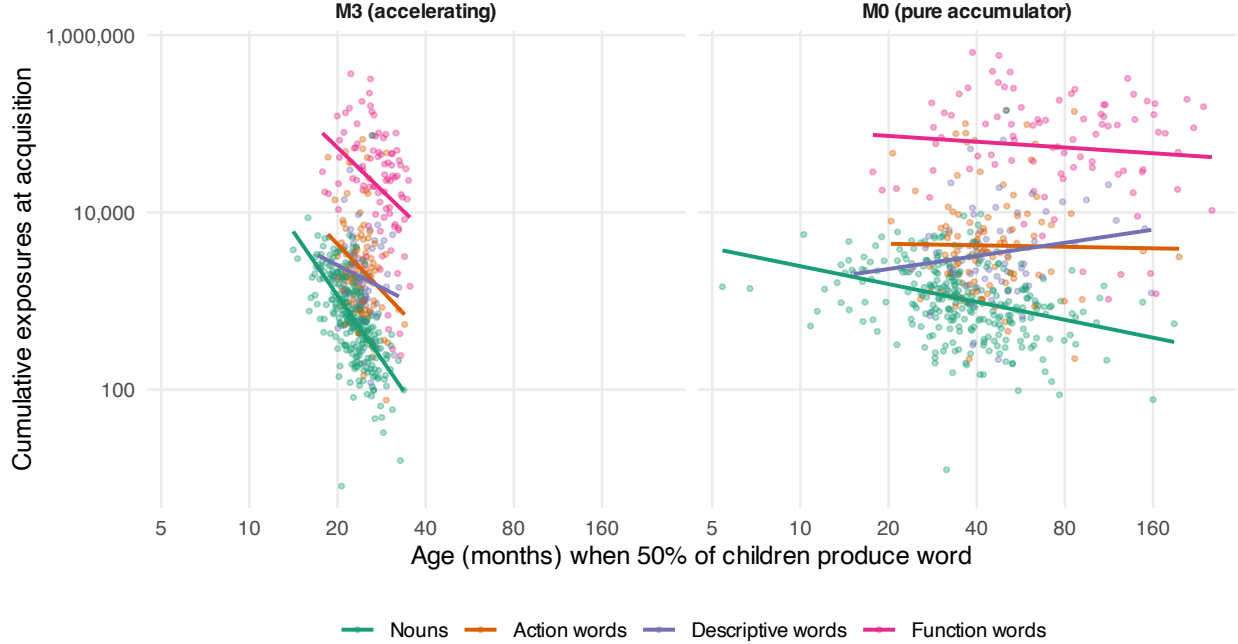


fig. S7: Estimated cumulative exposures at age of acquisition under the accelerating accumulator (M3, left, reproduced from main text) and the fitted pure accumulator (M0, right), using identical corpus frequencies. Points are words, colored by class, with per-class linear fits.

table S10: Slope of estimated cumulative exposures with respect to age of acquisition, by word class, under each item model. Negative values mean fewer exposures are needed for later-acquired words.

Word class	n	M3 (accelerating)		M0 (pure accumulator)	
		slope	R^2	slope	R^2
nouns	293	-4.77	0.38	-0.67	0.11
action words	103	-3.47	0.07	-0.06	0.00
descriptive words	62	-1.68	0.03	+0.49	0.03
function words	90	-3.18	0.13	-0.22	0.01

Computing Acceleration in LMs

To compare acceleration in LMs and children, we link the κ parameter from our accelerating accumulator model to the word-by-word learning performance of LMs. We first focus on the slope for individual words rather than individual children. Starting with the M3, differentiating $\eta_{ij}(t)$ (the probability of producing a particular word) with respect to $\log t$ gives κ_i : the logit-probability that a child produces a word rises linearly in log-age with slope equal to the child’s scaling exponent. Thus, the per-child sigmoid slope *is* κ_i , by construction.

Next, to estimate the slope of acquisition of individual words, (9) compute surprisal for each word across LM training checkpoints and then fit a four-parameter logistic, $s_w(x) = \ell_w + (u_w - \ell_w)/(1 + \exp((x - m_w)/\text{scale}_w))$, in $x = \log_{10}$ training data. Let $p_w = (s_w - \ell_w)/(u_w - \ell_w)$

be the fraction of that word’s surprisal reduction still outstanding. The LM slope is therefore $d \logit(p_w)/d \ln D = 1/(\text{scale}_w \ln 10)$ — a logit per unit of natural-log input. That is the same as κ_i for children, where the logit probability of producing a word rises in log-age with slope κ_i and the outstanding fraction is $1 - P(\text{produce})$.

Matched Estimation of Acceleration

One potential objection to the comparison of children and LMs in the main text is that the estimators are different in two ways. First, the estimator for children is Rasch-based, while the model from (9) is a four-parameter logistic. Second, and perhaps more saliently, the child estimator is for κ across children, while the LM estimator is across words.

To address these issues, we fit the LM four-parameter estimator directly to data on children’s word acquisition, averaging across children to produce a proportion of children producing each word at each age. (The Thal and Japanese datasets had fewer children and less age diversity, so not all words produced a viable curve estimate, but most words could be estimated for the Smith, Marchman, and Norwegian datasets). fig. S8 shows the results. Across all datasets, κ estimates were robustly higher than LMs, though somewhat attenuated relative to the M3 fits.

One possible source of this attenuation is the averaging of many different acquisition curves across children. To estimate this attenuation, we sampled from the fitted M3 parameters for ξ_i , κ_i and δ_j , and ran the identical pipeline over the data to recover a correction factor (the ratio of the estimated by-word κ to the population κ used to generate the data). We then applied this correction factor to our empirical estimates. table S11 shows these corrected estimates, which match very closely to the M3 values.

In sum, even with a completely matched estimator, acceleration diverges substantially between children and LMs.

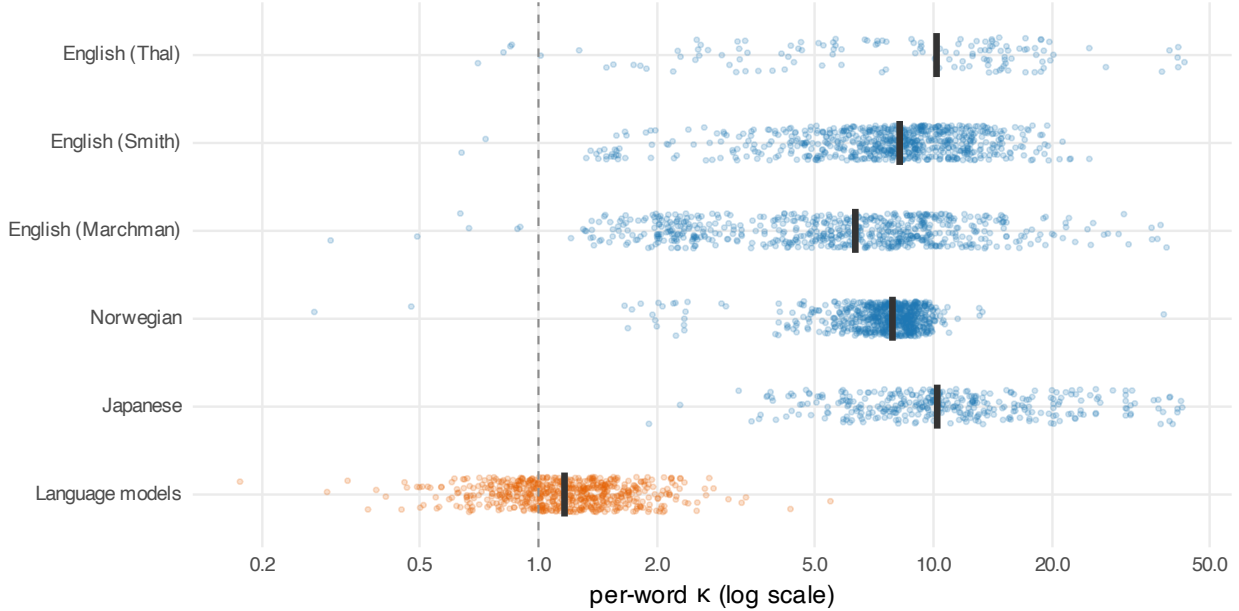


fig. S8: Per-word acceleration for children and language models, both estimated with the same four-parameter logistic. Points are words, with the median marked; language model values are medians across the ten training seeds. The dashed line at one is the pure-accumulator value. Child estimates are attenuated by pooling across children and so understate the separation; the correction factors are given in the table.

table S11: Per-word acceleration estimated with the language models’ own estimator. For each word we fit the same four-parameter logistic used on LM surprisal to the proportion of children not yet producing that word, and take $\kappa = 0.434/\text{scale}$. Recovery is the fraction of the population κ that the same pipeline returns when run on data generated from the fitted M3, and is the factor by which pooling across children attenuates the estimate; the corrected estimate is the raw estimate divided by this factor. The final column is the κ M3 model reports for that dataset.

Sample	words fitted	κ raw [IQR]	recovery	κ corrected	κ (M3)
English (Thal)	135/681	10.18 [4.32, 14.84]	98%	10.42	11.51
English (Smith)	673/680	8.21 [6.04, 10.67]	62%	13.21	12.95
English (Marchman)	624/681	6.34 [3.12, 9.29]	59%	10.66	10.65
Norwegian	713/733	7.87 [6.97, 8.69]	57%	13.81	13.25
Japanese	343/447	10.22 [7.66, 15.56]	86%	11.92	11.57
Language models	609/609	1.16 [0.93, 1.46]			

Other Architectures

The acceleration results we report in the main text are not specific to our choice of the GPT-2 architecture. We re-analyze the published fits of (9). fig. S9 shows a similar distribution of per-word κ values for the four different architectures reported in that study, which were trained on standard LM corpora rather than child-directed input.

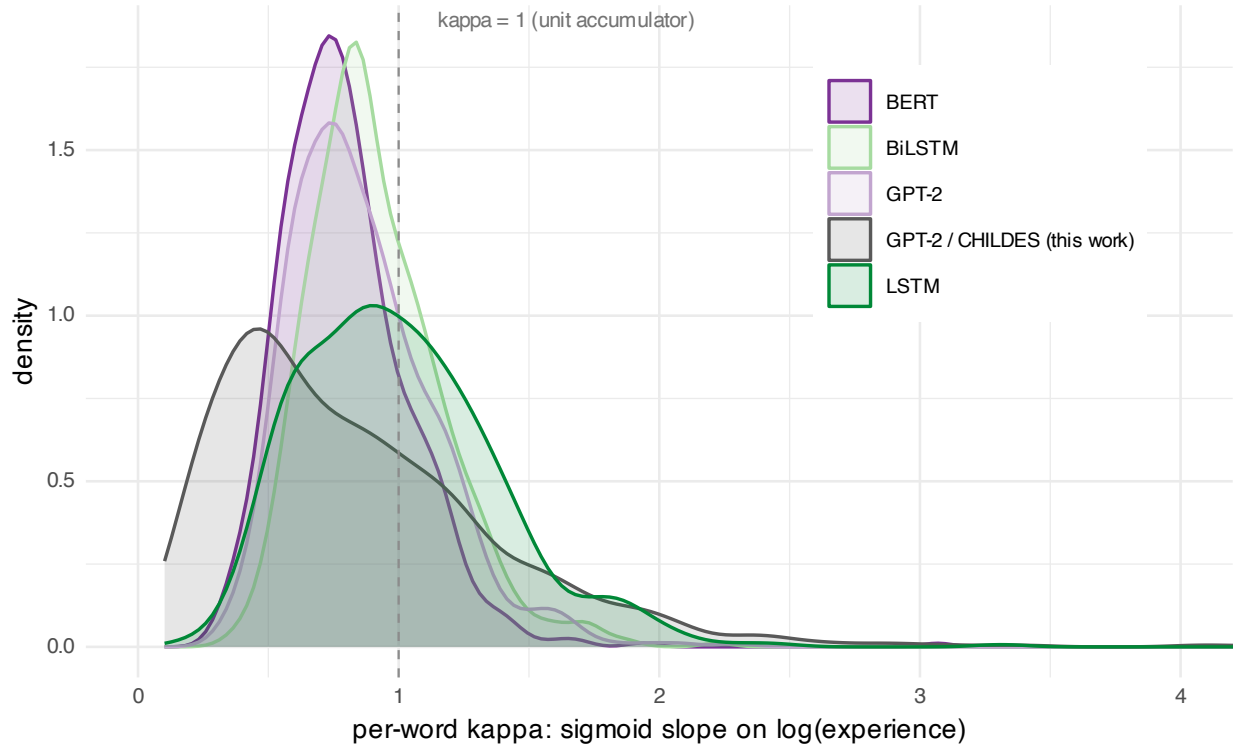


fig. S9: Per-word acquisition slopes for the four architectures studied in Chang & Bergen (2022) – BERT, GPT-2, BiLSTM, LSTM – trained on BookCorpus + WikiText-103. CHILDES-trained GPT-2 models (training axis) are overlaid in gray.

1. M. C. Frank, M. Braginsky, D. Yurovsky, V. A. Marchman, Wordbank: An open repository for developmental vocabulary data. *Journal of Child Language* **44**, 677–694 (2017).
2. A. Sanchez, S. C. Meylan, M. Braginsky, K. E. MacDonald, D. Yurovsky, M. C. Frank, Childes-db: A flexible and reproducible interface to the child language data exchange system. *Behavior Research Methods* **51**, 1928–1941 (2019).
3. B. Carpenter, A. Gelman, M. D. Hoffman, D. Lee, B. Goodrich, M. Betancourt, M. Brubaker, J. Guo, P. Li, A. Riddell, Stan: A probabilistic programming language. *Journal of statistical software* **76**, 1–32 (2017).
4. D. E. Sperry, L. L. Sperry, P. J. Miller, [Reexamining the verbal environments of children from different socioeconomic backgrounds](#). *Child Development* **90**, 1303–1318 (2019).
5. B. Hart, T. R. Risley, *Meaningful Differences in the Everyday Experience of Young American Children* (Brookes, 1995).

6. S. Y. Feng, N. D. Goodman, M. C. Frank, “Is child-directed speech effective training data for language models?” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, Y.-N. Chen, Eds. (Association for Computational Linguistics, Miami, Florida, USA, 2024; <https://aclanthology.org/2024.emnlp-main.1231/>), pp. 22055–22071.
7. M. Y. Hu, A. Mueller, C. Ross, A. Williams, T. Linzen, C. Zhuang, R. Cotterell, L. Choshen, A. Warstadt, E. G. Wilcox, “Findings of the second BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora” in *The 2nd BabyLM Challenge at the 28th Conference on Computational Natural Language Learning* (2024), pp. 1–21.
8. S. Diao, Y. Yang, Y. Fu, X. Dong, D. Su, M. Kliegl, Z. Chen, P. Belcak, Y. Suhara, H. Yin, M. Patwary, Y. Lin, J. Kautz, P. Molchanov, Nemotron-CLIMB: CLustering-based iterative data mixture bootstrapping for language model pre-training (2025). <https://arxiv.org/abs/2504.13161>.
9. T. A. Chang, B. K. Bergen, [Word acquisition in neural language models](#). *Transactions of the Association for Computational Linguistics* **10**, 1–16 (2022).
10. J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. Casas, L. A. Hendricks, J. Welbl, A. Clark, others, Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556* **10** (2022).
11. B. McMurray, Defusing the childhood vocabulary explosion. *Science* **317**, 631–631 (2007).
12. S. Hidaka, A computational model associating learning process, word attributes, and age of acquisition. *PloS one* **8** (2013).
13. G. Kachergis, V. A. Marchman, M. C. Frank, Toward a “standard model” of early language learning. *Current Directions in Psychological Science* **31**, 20–27 (2022).
14. C. Mitchell, B. McMurray, On leveraged learning in lexical acquisition and its relationship to acceleration. *Cognitive Science* **33**, 1503–1523 (2009).
15. F. Mollica, S. T. Piantadosi, How data drive early word learning: A cross-linguistic waiting time analysis. *Open Mind* **1**, 67–77 (2017).
16. V. A. Marchman, P. Dale, L. Fenson, *MacArthur-Bates Communicative Development Inventories User’s Guide and Technical Manual, Third Edition* (Brookes Publishing, 2021).
17. G. Kachergis, V. A. Marchman, P. S. Dale, J. Mankewitz, M. C. Frank, Online computerized adaptive tests of children’s vocabulary development in english and mexican spanish. *Journal of Speech, Language, and Hearing Research* **65**, 2288–2308 (2022).

18. K. A. Adams, V. A. Marchman, E. C. Loi, M. D. Ashland, A. Fernald, H. M. Feldman, Caregiver talk and medical risk as predictors of language outcomes in full term and preterm toddlers. *Child Development* **89**, 1674–1690 (2018).
19. B. Long, R. Z. Sparks, V. Xiang, S. Stojanov, Z. Yin, G. Keene, A. W. M. Tan, S. Y. Feng, A. Nag, C. Zhuang, V. A. Marchman, D. L. K. Yamins, M. C. Frank, “[The BabyView dataset: High-resolution egocentric videos of infants’ and young children’s everyday experiences](#)” in *Proceedings of the Conference on Cognitive Computational Neuroscience* (2025).
20. A. Fernald, V. A. Marchman, A. Weisleder, SES differences in language processing skill and vocabulary are evident at 18 months. *Developmental science* **16**, 234–248 (2013).
21. S. Egan-Dailey, E. Bergelson, Early child measures outpredict input measures of preschool language skills in US english learners. *Developmental psychology* (2025).